		Organisation's SANAS No/s.
sanas South African National Accreditation System	**VERTICAL ASSESSMENT ISO/IEC 17025:2017 LABORATORIES**	

Date/s of Assessment	29/01 – 20/02 2026
Assessor/s & Observers	Iliso Myambi
Laboratory	Mzuvelela Pty Ltd.
Area / field of operation	Testing
Laboratory Representative	Keitumetse Malcka, Lulo Ndlebe, Nosiplo Hlongwane & Abonathi Jama

Certificate or Report *(Select one or more final Report /Certificate(s). Record at least the number, date and the accredited parameters, as on the Scope of Accreditation, measured. Include a short description of calibrations/tests performed,*

AZ PT 001

Name of analyst / metrologist *and* technical signatory

Iliso Myambi

Clause	**REQUIREMENTS AND COMMENTS** *Compliance = C, Non-compliance = NC, Not applicable = NA* ***NB1: Indicate WHAT has been checked and HOW requirements have been implemented. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this checklist is designed as guidance to prompt detailed recording of the process. NB2*: Where a clause is marked as NA, reason must be provided as to why it's not applicable REFER TO ISO/IEC 17025:2017 FOR DETAIL AND FOR CLARIFICATION NOTES.***	C NC NA*
7.5 / 7.11	**Technical Records** *(state which data and calculations were checked)*	
7.5.1	Records include the date and the identity of personnel responsible	C
7.5.1	Records contain sufficient information	C
7.5.2	Amendments to technical records traceable to previous versions or to original observations	C
7.5.2	Original, amended data and files are kept, including the date of alteration and person responsible	C
7.11.1	Access to the data and information needed to perform laboratory activities	C
7.11.2	LIMS validated for functionality before introduction	C
7.11.6	Calculations and data transfers checked in an appropriate and systematic manner	C

6	**Personnel - Training** *Check training records of identified operator(s) for accredited work from selected Results*	
6.2	Operator/s identified as competent for the work and is proof of competence available	C
6.2.2	Personnel competence documented for each activity	C
6.2.3	Personnel competent for the work and able to evaluate the significance of deviations	C
6.2.4	Duties, responsibility and authority communited to personnel	C
6.2.5	Records retained for: training; supervision; authorisation; monitoring of competence	C
6.2.6	Personnel authorized to perform specific function	C

Additional Assessor Notes *(This may be used for rough notes as well)*

7.2.1-3/ 7.6	**Performance capability of selected methods**	
7.2.1.1 - 7	Proof of confirmation of proper operation of -standard methods, -laboratory developed methods -non-standard methods	C C C
7.2.2.1 - 4	Methods validated and availability of performance capability Modifications to the validation development plan approved and authorized. Validation records retained (procedure; requirements; performance characteristics; results & fitness for purpose statement)	C C
7.6.1	Contributions to measurement uncertainty identified	C
7.6.2	Calibration laboratories (including in-house calibration) - evaluate the measurement uncertainty for all calibrations	C
7.6.3 ***Notes Std***	Testing laboratories - evaluate measurement uncertainty.	C

7.7	**Ensuring the validity of results**	
7.7.1 (a – k)	Indicate how the laboratory monitors results Monitoring data suitably recorded (e.g. control charts), evaluated reviewed.	C
7.7.2	Monitor its performance by comparison with results of other laboratories	C
7.7.3	Monitoring data analysed, used to control and/or improve the laboratory's activities Action taken for incorrect results or results found outside pre-defined criteria	C

6.4 / 6.5	**Calibration of equipment and/or standards used** *(Indicate which equipment/standards)*	
6.4.6	Measuring equipment are calibrated	C
6.4.7	Appropriateness of calibration and verification programmes, cover operating ranges	C
6.4.8	Calibrated equipment are correctly labelled, coded or identified	C
6.4.11	Calibration and reference material data are updated and implemented	C
6.4.13	Records of calibration and verification complete, tolerances appropriate	C
6.5.1	Establish and maintain metrological traceability of its measurement results	C
6.5.2	Traceability to national standards	C
6.5.3	Metrological traceability to an appropriate reference e.g. CRMs, Consensus standards	C
6.4.10	Intermediated checks - reference, primary, transfer and working standards	C

Additional Assessor Notes *(This may be used for rough notes as well)*

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6.4	**Equipment maintenance and operation**	
6.4.3 6.4.4 6.4.9 6.4.12	Handling/transport/storage/use to prevent contamination/unintended adjustment/deterioration of - equipment and - std/reference materials	C
6.4.13 a-h	Records for equipment which can influence laboratory activities retained	C

7.8	**Test Report / Certificate**	
7.8.1.1 / R03	Results reviewed and authorized prior to release	C
7.8.1.2	Simplified report issued with the agreement of the customer	C
7.8.2.1	Report/certificate include at least following information, unless the laboratory has valid reasons:	
a)	a title (e.g. "Test Report" or "Calibration Certificate").	C
b)	name and address of the laboratory	C
c)	the location of performance of the laboratory activity.	C
d)	unique identification	C
e)	the name and contact information of the customer	C
f)	identification of the method used	C
g)	a description, unambiguous identification, and, when necessary, the condition of the item	C
h)	the date of receipt of the test or calibration item(s), and the date of sampling, where this is critical to the validity and application of the results;	C
i)	the date(s) of performance of the laboratory activity;	C
j)	the date of issue of the report;	C
k)	reference to the sampling plan and sampling method used by the laboratory or other bodies where these are relevant to the validity or application of the results;	
l)	statement to the effect that the results relate only to the items tested, calibrated or sampled;	C
m)	the results with, where appropriate, the units of measurement;	C
n)	additions to, deviations, or exclusions from the method;	
o)	identification of the person(s) authorizing the report	C
p)	clear identification when results are from external providers	C
7.8.2.2	Data provided by a customer clearly identified	C
	A disclaimer is put on the report when the information is supplied by the customer and can affect the validity of results	C
	If lab is not responsible for the sampling stage, it is stated in the report that the results apply to the sample as received	
R04	Correct use of the SANAS Accreditation Symbol (SANAS R04)	C

The lab is currently not accredited.

7.8.3	**Specific requirements for test reports**	
7.8.3.1	In addition to 7.8.2, do test reports, where necessary for Interpretation of test results, include:	
a)	information on specific test conditions, such as environmental conditions	
b)	where relevant, a statement of conformity with requirements or specifications	
c)	where applicable, the measurement uncertainty presented in the same unit as that of the measurand or in a term relative to the measurand	
d)	where appropriate, opinions and interpretations (see 7.8.7);	
e)	additional information which may be required by specific methods, authorities, customers or groups of customers	
7.8.4	**Specific requirements for calibration certificates**	
7.8.4.1	In addition to the requirements listed in 7.8.2, calibration certificates shall include the following	
a)	the measurement uncertainty of the measurement result presented in the same unit as that of the measurand or in a term relative to the measurand	C
b)	the conditions (e.g. environmental) under which the calibrations were made that have an influence on the measurement results	C
c)	a statement identifying how the measurements are metrologically traceable	C
d)	the results before and after any adjustment or repair, if available	C
e)	where relevant, a statement of conformity with requirements or specifications	
f)	where appropriate, opinions and interpretations	
7.8.4.2	Is the laboratory responsible for sampling? NO.	
7.8.4.3	Calibration certificate or calibration label shall not contain any recommendation on the calibration interval except where this has been agreed with the customer.	
	7.8.4.1- Applies to certificates given by the calibrating lab.	

7.8.5	**Reporting sampling – specific requirements**	
Where the laboratory is responsible for the sampling activity, in addition to the requirements listed in 7.8.2, reports shall include the following, where necessary for the interpretation of results:		
a)	the date of sampling	C
b)	unique identification of the item or material sampled (including the name of the manufacturer, the model or type of designation and serial numbers as appropriate)	C
c)	the location of sampling, including any diagrams, sketches or photographs	C
d)	a reference to the sampling plan and sampling method	C
e)	details of any environmental conditions during sampling that affect the interpretation of the test results	C
f)	information required to evaluate measurement uncertainty for subsequent testing or calibration	N/A

The lab hasn't sampled any of its samples. However, the above details have been included in the report structure for future endeavours.

Additional Assessor Notes *(This may be used for rough notes as well)*

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7.8.6	**Reporting statements of conformity**	
7.8.6.1	Decision rule documented when statement of conformity to a specification or standard is provided	
7.8.6.2	Statement of conformity clearly identifies:	
a)	to which results the statement of conformity applies	
b)	which specifications, standards or parts thereof are met or not met	
c)	the decision rule applied (unless it is inherent in the requested specification or standard).	
7.8.7	**Reporting opinions and interpretations**	
7.8.7.1 Notes Std	Only authorised personnel express opinions and interpretations	
7.8.7.2	Opinions and interpretations expressed are based on the results obtained from the tested or calibrated item and clearly identified	
7.8.7.3	Record of the dialogue with the customer retained	
7.8.8	**Amendments to reports.**	
	Any change of information clearly identified and, where appropriate, the reason for for the change included in the report	C
	Amendments to a report after issue only made in the form of a further document, or data transfer which includes the statement "Amendment to Report, serial number or an equivalent form of wording	C
	A complete new report is uniquely identified and contain a reference to the original that it replaces	C

for 7.8.8. see report AZ PT 006.

6.3	**Facilities and environmental conditions**	
6.3.1 ***Notes Std***	Suitability of facilities and environmental conditions for the laboratory activities	C
6.3.2	Requirements for facilities and environmental conditions documented	C
6.3.3	Environmental conditions monitored, controlled and recorded	C
6.3.4	Measures for access control, cross-contamination prevention and effective separation implemented; monitored and reviewed periodically	C
6.3.5	Suitability of facilities or sites outside laboratory's permanent control	C

Additional Assessor Notes (This may be used for rough notes as well)

Signed: Technical Assessor			
Signed : Team Leader		Date	

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SANAS Vertical Assessment Audit Checklist

Discipline: Coal / Solid Fuels Analysis
Methods:
☐ Ash (e.g. ISO 1171)
☐ Volatile Matter (e.g. ISO 562)
☐ Inherent Moisture (e.g. ISO 589)
☐ Calorific Value (e.g. ISO 1928)

Sample / Job No: _____
Analyst: _____
Date: 29/01/2026.
Auditor: Sifiso H d Thlepo M.

1. SAMPLE RECEIPT & IDENTIFICATION

Clauses: 7.4, 7.5, 8.4

Requirement	Yes	No	N/A	Evidence / Comments
Sample uniquely identified on receipt	☑	☐	☐	⇒See receiving log Bk.
Condition of sample recorded (wet/damaged/insufficient)	☑	☐	☐	Added 29/01/2026
Chain of custody / job card traceable	☐	☐	☐	⇒Added 29/01/2026
Sample storage conditions controlled	☑	☐	☐	⇒Technical records.
Sample handling instructions available	☑	☐	☐	⇒Preparation procedure

2. SAMPLING (if laboratory performs sampling)

Clauses: 7.3, 7.8.5

Requirement	Yes	No	N/A	Evidence / Comments
Sampling plan documented	☑	☐	☐	Added 29/01/2026
Statistical basis for representativeness defined	☑	☐	☐	29/01/2026
Sampling location recorded	☑	☐	☐	29/01/2026
Deviations from plan recorded	☑	☐	☐	29/01/2026
Environmental conditions recorded (where relevant)	☑	☐	☐	29/01/2026

Sampling Procedure ✓

3. SAMPLE PREPARATION (CRUSHING / PULVERISING)

Clauses: 6.4, 7.2, 7.5

Requirement	Yes	No	N/A	Evidence / Comments
Sample preparation SOP available	☑	☐	☐	
Crusher/pulveriser uniquely identified	☑	☐	☐	
PSD (fineness) verification performed	☐	☐	☐	
Acceptance criteria defined (e.g. >95% passing 212 µm)	☐	☐	☐	
Cleaning between samples recorded	☐	☐	☐	
Maintenance records available	☐	☐	☐	

4. METHOD CONTROL (ASH, VM, IM, CV)

Clauses: 7.2

Requirement	Yes	No	N/A	Evidence / Comments
Standard methods identified (ISO/ASTM/SANS)	☑	☐	☐	⇒ ISO standards linked to procedures.
Methods authorized for use	☑	☐	☐	
Method verification performed	☑	☐	☐	
Method deviations documented	☐	☐	☐	
Working instructions available at point of use	☑	☐	☐	

5. PERSONNEL COMPETENCE

Clauses: 6.2

Requirement	Yes	No	N/A	Evidence / Comments
Analyst authorized for Ash	☑	☐	☐	
Analyst authorized for VM	☑	☐	☐	
Analyst authorized for IM	☑	☐	☐	
Analyst authorized for CV	☑	☐	☐	

Requirement	Yes	No	N/A	Evidence / Comments
Training records available	☑	☐	☐	

6. TECHNICAL RECORDS & RAW DATA

Clauses: 7.5

Requirement	Yes	No	N/A	Evidence / Comments
Raw observations recorded at time of test	☑	☐	☐	
Analyst identified on worksheet	☑	☐	☐	
Environmental conditions recorded (if required)	☐	☐	☐	
Corrections traceable (initial/date)	☐	☐	☐	
Data traceable to sample ID	☑	☐	☐	

7. EQUIPMENT CONTROL

Clauses: 6.4

7.1 Furnaces (Ash & VM)

Requirement	Yes	No	N/A	Evidence / Comments
Furnace uniquely identified	☑	☐	☐	
Temperature mapping performed	☐	☐	☑	
Hot zone defined	☐	☐	☑	
Verification frequency defined	☑	☐	☐	
Daily temperature checks recorded	☐	☐	☑	

7.2 Balances

Requirement	Yes	No	N/A	Evidence / Comments
Balance uniquely identified	☑	☐	☐	
Calibration certificate available	☑	☐	☐	

Requirement	Yes	No	N/A	Evidence / Comments
Daily performance checks recorded	☑	☐	☐	
Out-of-tolerance procedure defined	☑	☐	☐	

7.3 Bomb Calorimeter

Requirement	Yes	No	N/A	Evidence / Comments
Energy equivalent verified	☑	☐	☐	
Benzoic acid calibration performed	☑	☐	☐	
Corrections applied (acid, fuse wire)	☐	☐	☑	
Calibration frequency defined	☑	☐	☐	

8. METROLOGICAL TRACEABILITY

Clauses: 6.5

Requirement	Yes	No	N/A	Evidence / Comments
Calibration traceable to SI units	☑	☐	☐	
SANAS-accredited certificates used	☑	☐	☐	
Uncertainty stated on certificates	☑	☐	☐	
Reference materials traceable	☑	☐	☐	

9. QUALITY CONTROL / VALIDITY OF RESULTS

Clauses: 7.7

Requirement	Yes	No	N/A	Evidence / Comments
CRMs used for Ash	☑	☐	☐	

Requirement	Yes	No	N/A	Evidence / Comments
CRMs used for VM	☑	☐	☐	
CRMs used for IM	☑	☐	☐	
CRMs used for CV	☑	☐	☐	
Control charts maintained	☑	☐	☐	
Control limits defined	☑	☐	☐	
Trends reviewed	☑	☐	☐	
Actions taken on failures	☑	☐	☐	

10. CALCULATIONS & DATA TRANSFER

Clauses: 7.11, 7.5

Requirement	Yes	No	N/A	Evidence / Comments
Correct formulas used	☑	☐	☐	
Dry/air-dry basis correctly applied	☑	☐	☐	
Data transfer checked	☑	☐	☐	
Spreadsheet validated	☑	☐	☐	

11. NONCONFORMING WORK

Clauses: 7.10

Requirement	Yes	No	N/A	Evidence / Comments
QC failures recorded	☑	☐	☐	
Root cause analysis performed	☑	☐	☐	
Impact on results evaluated	☑	☐	☐	
Corrective action implemented	☑	☐	☐	

12. REPORTING OF RESULTS

Clauses: 7.8

Requirement	Yes	No	N/A	Evidence / Comments
Unique report number	☑	☐	☐	
Client and sample ID stated	☑	☐	☐	
Methods stated	☑	☐	☐	
Units correct (%, MJ/kg)	☑	☐	☐	
Date of issue	☑	☐	☐	
Authorization signature	☑	☐	☐	
Sampling disclaimer (if applicable)	☑	☐	☐	
Amendments controlled	☑	☐	☐	

13. OVERALL ASSESSMENT

Conformity Summary:

☑ Sample traceable
☑ Method valid
☑ Equipment fit for purpose
☑ Analyst competent
☑ QC supports result
☑ Report compliant

Nonconformities Raised:

Auditor Signature: _____
Date: 29/01/26

	TECHNICAL SIGNATORY COMPETENCE DECLARATION FORM
	Document no:

Equipment name	Company no.	Certificate pres d match equip no	Verification records	Efficiency records.	Frequency defined?	Deviations/ comments	
Inherent moisture OV	AS3800	Yes	Present	NA	Yes	Thermocouple no not defined	
Vol Matter furn	AS01338	Yes	Present	NA	Yes	Ave+ corr factor not applied.	
Ash furn.	AS02217	Yes	Present	NA	Yes	Ave+ corr factor not applied.	
Hygrometer	AS0096D	Yes, but certificate no does not match	NA	NA	Yes	OBS TH 3.1	
Pulveriser	AS021PA (not clearly visible)	NA	~~Present~~/ NA	Present	Yes		
Pulveriser	AS7WET4	NA	NA	Present	Yes	Aug Records Jul not present (sent for maintenance)	
Pulveriser	NA	NA	NA	NA	NA	NOT in use (spare)	
Platform	AS025PA	Yes.	Yes	NA	Yes.		
Analytical	AS2156	Yes	December records not there NC.	NA	Yes.		
Top pan	AS00A3	Yes	~~Dec no records~~ Yes	N/A	Yes		
Calorific Value.	AS022PA	NA	NA 23/09/25 w/o NC.	N/A	Yes.		
Crusher -25mm	AS8284	NA	Yes	Present	Yes		
-10mm	AS8809	NA	Yes	Present	Yes		
- 3mm	AS5935	NA	Yes	Present	Yes		

Parameter/Item	Abenathi	Nosipho	Kretumelo	Lilo
AR	Done & signed	Done & signed	Done & signed	Done & signed.
CV	Done but not signed (NC)	Done & signed	Done & signed	Done & signed.
VBI Matter	Done & signed	Done & signed.	D & S	D & S.
Pub. Moist	D & S	D & S	D & S	D & S
Witnessing	D & S	D & S	D & S	D & S.
Sample prep	Done but not witnessed	Done but not W.	D & W.	D & W.
LIMS number on competency traceable to system?	Yes. VBI corrected	Yes. VBI correct	Yes. VBI corrected	Yes. VBI corrected.

6. Technical Records & Raw data

Records for environmental conditions

Aug

Sept

Oct Not controlled document, Supervisor's signature not there. 30/07

Data Traceable to sample ID?

Humidity is out of spec in some of the days but analysts continued to work.

Quality Control / Validity of results

CRM used for ash ✓	CRM used for VM ✓	CRM used for IM ✓
Control charts present ✓	Control charts present ✓	
limits defined.	limits defined	limits defined ✓

Calculations / limits verification

	IM	VG	ASH
Nosipho	NC.	✓	✓
Lilo	✓	✓	✓
Abenathi	✓	✓	✓
Keitumetse	✓	✓	✓